

1.9 Adding and Subtracting Radical Expressions

Lesson Introduction

 Benchmark	MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.			 Learning Target	I can add or subtract two square roots or two cube roots.
 Benchmark Coherence	Building On			Working Toward	
	MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.			MA.912.NSO.1.5 Add, subtract, multiply, and divide algebraic expressions involving radicals.	
 Essential Questions	- How do I add or subtract two radical expressions? - What is the process for determining if two radical expressions can be added?				
 Lesson Narrative	In this lesson, students will extend their knowledge of radical expressions to add and subtract them. Students will experience situations where the radical expression contains the same radical values and situations where they contain different radical values.				
 Component Summary	1.9.1 Warm-Up (~5 min.)	Adding and Subtracting Algebraic Expressions Activity (<i>SE p. 45</i>)	1.9.4 Guided Practice (5 min.)	Adding and subtracting radical expressions practice activity (<i>SE p. 48</i>)	
	1.9.2 Exploration (5 min.)	Exploring the addition and subtraction of radical expressions (<i>SE p. 46</i>)	1.9.5 Your Turn! (5 min.)	Independent practice finding the sum and/or difference of radical expressions (<i>SE p. 49</i>)	
	1.9.3 Guided Instruction (20 min.)	Adding and subtracting radical expressions instructional activity (<i>SE p. 46</i>)	1.9.6 Wrap-Up (~5 min.)	Ticket Out the Door on finding the sum of a radical expression (<i>SE p. 49</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Like Radicals <u>Additional Terms:</u> - Rational Exponent - Radical Form - Equivalent Expression		N/A		Consider potential interventions that may be necessary based on a student's unfinished learning from a prior lesson or prior grade level. For example, see the differentiation note for the 1.9.1 Warm-Up activity. If students struggle with question 1, they may need intervention related to combining like terms.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not realize that a radical can be simplified in order to be added. - Students may make calculation errors when adding or subtracting. - Students may not know their factor pairs well enough to simplify the radicals.	This lesson incorporates several “select all that apply” style questions. Consider reminding students that these types of questions can appear on high-stakes tests and encourage students to critically think through each answer choice prior to completing the entire question. “Select all” questions are often a good way to assess fluency and/or equivalency.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	True or False In 1.9.4 Guided Practice, students will choose all the expressions that could be combined. Consider leveraging a True or False routine for this activity. Present each answer choice and state that they can be combined. Have students determine if that statement is true or if it is false, discussing why it is either answer.	MA.K12.MTR.4.1: Discussion and Reflection In 1.9.2 Exploration, students will review the work of two students as they add two rational expressions. Students will determine which student has made an error and describe how to correctly complete the task.
 Differentiate Instruction	Consider partnering students up based on their performance in the prior lesson, as students will have different entry points to this lesson. Consider having a high-medium and medium-low pairing scenario where students who performed well with the previous lesson are paired with those who did not. The knowledge of the prior lesson is essential to success in this one. 1.9.3 Guided Instruction provides students with three steps to be used when adding and subtracting rational expressions. Consider creating a visual aid such as an anchor chart to post these three steps for students to reference. On the visual aid, show at least three examples of radical expressions being added and subtracted following each of these three steps. Students will complete a Ticket Out the Door that assesses student understanding of adding and subtracting radical expressions. Student performance on this activity can provide data for potential differentiation that may be needed for specific students.	
Lesson Notes:		

1.9.1 Warm-Up: Adding and Subtracting Algebraic Expressions

Component Overview: Students will complete a series of questions that have them combine like terms provided in different forms, then students make a prediction about like terms of radical expressions that will be explored during the lesson.



1.9.1 Warm-Up: Adding and Subtracting Algebraic Expressions

1. Find the sum or difference of each algebraic expression.

$$4x + 23x$$

$27x$

$$-7t - 40t$$

$-47t$

$$-35b + 10b$$

$-25b$

2. Complete the statement to explain why it is possible to combine the terms in each of these expressions.

When adding and subtracting algebraic expressions, **like** terms can be combined.

3. Which, if any, of the radical expressions do you think can be combined?

- ☐ $\sqrt{17} + \sqrt{32}$
- ☐ $5\sqrt{13} - 8\sqrt{3}$
- ☒ $-7\sqrt{17} + 3\sqrt{17}$
- ☒ $9\sqrt{5} + \sqrt{5}$
- ☐ $8\sqrt{40} - 4\sqrt{9}$
- ☒ $11\sqrt{48} - 6\sqrt{48}$
- ☐ $-2\sqrt{49} - 3\sqrt{4}$



Watch for student performance on questions 1 and 2. If students are struggling to get the correct answers for question 1, there may be need for additional prerequisite intervention. Understanding the concept of like terms and being able to combine them appropriately will be a key skill for adding and subtracting radical expressions.



- How did you determine which radical expressions in question 3 could be combined?
- What is the relationship between what you know about like terms and the expressions in question 3?

1.9.2 Exploration: Adding and Subtracting Radical Expressions

Component Overview: Students will work in small groups to explore the addition and subtraction of radical expressions. For this activity, release students to complete questions 1 – 4 with their small groups. Bring the class back together and go through each question as an entire class.



1.9.2 Exploration: Adding and Subtracting Radical Expressions

1. In small groups, determine the value of each expression.

$$\sqrt{25} + \sqrt{100}$$

15

$$\sqrt{25 + 100}$$

11.1803

2. Explain which of the expressions is equivalent to $\sqrt{125}$.
The value of $\sqrt{25 + 100}$ is equivalent to $\sqrt{125}$ because the values under the radicand must be combined first.

3. Paloma says $\sqrt{16} - \sqrt{9} = \sqrt{7}$. Ty'rique says $\sqrt{16} - \sqrt{9} = 1$. Who is correct? Verify by entering $\sqrt{16} - \sqrt{9}$ into a calculator.

Ty'rique is correct because $\sqrt{16} - \sqrt{9}$ is equal to 1.

4. Xavier says $\sqrt{12} + 2\sqrt{12} = 3\sqrt{12}$. Sam says $\sqrt{12} + 2\sqrt{12} = 2\sqrt{24}$. Who is correct? Verify using a calculator.

Xavier is correct because $\sqrt{12} + 2\sqrt{12}$ is equal to $3\sqrt{12}$.



Students will need to do verification of values of radical expressions. Some students may need a refresher of how to use their calculator to do this appropriately. Consider using an on-screen calculator or document camera to show the steps to evaluating a radical expression.



Once the activity is completed, allow students to share out their responses and methods for each question. Consider polling the class to determine how many students arrived at the same conclusion. Have students vote on which students they believe were correct in questions 3 and 4, sharing their reasoning for their answer.



When verifying using a calculator, what do you notice when taking the square root of 7 vs taking the square root of 16 and subtracting the square root of 9?

1.9.3 Guided Instruction: Adding and Subtracting Radical Expressions

Component Overview: Students will receive guided instruction on how to add and subtract radical expressions using like radicals and rewriting radical expressions in order to add or subtract like radicals.



1.9.3 Guided Instruction: Adding and Subtracting Radical Expressions

When adding and subtracting algebraic expressions, only **like terms** can be combined. Adding and subtracting radical expressions is similar because only like radicals can be combined.



Like radicals are radical terms that have the same index and the same radicand.

1. Find the sum or difference of each radical expression.

a. $4\sqrt{5} + 7\sqrt{5} = 11\sqrt{5}$

b. $30\sqrt[3]{2} - 23\sqrt[3]{2} = 7\sqrt[3]{2}$

c. $-10\sqrt{7} - 23\sqrt{7} = -10\sqrt{7} - 23\sqrt{7}$

d. $-\sqrt{17} + \sqrt{17} = 0$

Three steps can be used when adding and subtracting radical expressions.

Step 1: Find perfect squares (for square roots) or perfect cubes (for cube roots) that are factors of the radicand.

Step 2: Rewrite the radical expressions by applying the product law of radicals and evaluating any perfect squares or cubes.

Step 3: Add or subtract like radical terms.



- What is the relationship between each of these radicals in question 1?
- Can you work with like radicals the same way you can with like terms? Why or why not?



Consider creating a visual aid such as an anchor chart to post these three steps for students to reference. On the visual aid, show at least three examples of radical expressions being added and subtracted following each of these three steps.

1.9.3 Guided Instruction: Adding and Subtracting Radical Expressions (continued)

2. Use the steps to add the following radical expression.

$$3\sqrt{6} + 5\sqrt{24}$$

Step 1: Both terms include square roots. If possible, find perfect squares that are factors of the radicands, 6 and 24.

$$6 = 3 \cdot 2$$

$$24 = 6 \cdot 4$$

Step 2: Rewrite the radical expressions by applying the product law of radicals and evaluating any perfect squares.

$$3\sqrt{6} + 5\sqrt{24} = 3\sqrt{6} + 10\sqrt{6}$$

Step 3: Add or subtract like radical terms.

$$13\sqrt{6}$$

3. Three students added $\sqrt{180}$ and $\sqrt{45}$ differently. Their work is shown.

Gerald's Work	Kalo's Work	Paola's Work
$\sqrt{180} + \sqrt{45}$	$\sqrt{180} + \sqrt{45}$	$\sqrt{180} + \sqrt{45}$
$2\sqrt{45} + \sqrt{45}$	$3\sqrt{20} + 3\sqrt{5}$	$6\sqrt{5} + 3\sqrt{5}$
$3\sqrt{45}$		$9\sqrt{5}$

- a. Use a calculator to verify if the students' answers are equivalent.

All three are equivalent.

- b. Why did the students end up with different expressions for their sum?

The students chose different factors.



When guiding students through question 2, be sure to teach each step explicitly, showing students how that step is being applied to this specific question. Have students discuss why each step is necessary in this problem and how to verify the step is done correctly.



Explain if steps 1 and 2 are necessary for every situation where you must add or subtract rational expressions.



Consider having students share their methods for question 3a. Use an on-screen calculator or a document camera to have students use their calculator to walk the class through the steps they used to verify each student's work.

1.9.3 Guided Instruction: Adding and Subtracting Radical Expressions (continued)

- c. Show how Gerald and Kalo could continue their work to arrive at the same expression as Paola.

Gerald's Next Steps	Kalo's Next Steps
$3\sqrt{45}$	$3\sqrt{20} + 3\sqrt{5}$
$3 \cdot \sqrt{9} \cdot \sqrt{5}$	$3 \cdot \sqrt{4} \cdot \sqrt{5} + 3\sqrt{5}$
$3 \cdot 3 \cdot \sqrt{5}$	$3 \cdot 2 \cdot \sqrt{5} + 3\sqrt{5}$
$9 \cdot \sqrt{5}$	$6 \cdot \sqrt{5} + 3\sqrt{5}$
$9\sqrt{5}$	$6\sqrt{5} + 3\sqrt{5}$
	$9\sqrt{5}$

4. Megh showed her work on the problem.

$$\sqrt[3]{32} + \sqrt[3]{64} = \sqrt[3]{8 \cdot 4} + 4 = 2\sqrt[3]{4} + 4 = 6\sqrt[3]{4}$$

Write an explanation to Megh explaining the error in her work.

The final answer should have been left at $2\sqrt[3]{4} + 4$. These terms are not like terms and can not be combined.



Have students share out their answer to question 3c. Consider having a student come to the board for each student and showing how to complete the remainder of the steps to arrive at the same answers as Paola. Have students not at the board offer feedback and support for the student at the board.

As students gain in their proficiency, release the students from showing work. Students do not have to show all the steps that are shown in the key.



- What makes the expression in question 4 different from the others that you have experienced?
- What about the process, if anything, was different from the other expressions?

1.9.4 Guided Practice: Adding and Subtracting Radical Expressions

Component Overview: Students will practice adding and subtracting radical expressions in a structured format. For this activity, release students to work with a partner to answer each question.



1.9.4 Guided Practice: Adding and Subtracting Radical Expressions

1. Select all of the expressions that can be combined. For the selected expressions, combine the terms. Show your work.

☐ $\sqrt{17} + \sqrt{32}$

☐ $5\sqrt{13} - 8\sqrt{3}$

☒ $-7\sqrt{17} + 3\sqrt{17}$

☒ $9\sqrt{5} + \sqrt{5}$

☐ $8\sqrt{40} - 4\sqrt{9}$

☒ $11\sqrt{48} - 6\sqrt{48}$

☒ $-2\sqrt{49} - 3\sqrt{4}$

With a partner, find the sum and difference of the following problems. If you cannot add or subtract, circle the expression.

a. $\sqrt[3]{6} + 3\sqrt{6}$

c. $16^{\frac{1}{3}} - 2^{\frac{1}{3}}$
 $2^{\frac{3}{2}} - \sqrt[3]{2} = \sqrt[3]{2}$

b. $\sqrt{20} - 3\sqrt{45}$
 $-7\sqrt{5}$

d. $\sqrt{72} + \sqrt{18} - \sqrt{15}$
 $9\sqrt{2} - \sqrt{15}$



Remind students that this is an example of a “select all that apply” question. Students should go through each answer choice to determine if it can be combined using what they’ve learned in this lesson.



When reviewing the answers, consider leveraging a True or False protocol. Present each answer choice and state that they can be combined. Have students determine if that statement is true or if it is false, discussing why it is either answer.



For a challenge, have students create a three-term radical expression that, when simplified, will reduce to a single-term radical expression. Have them show their work to prove that this will happen. Also, consider having students trade their expression with a partner, and have them evaluate to determine if it reduces to a single term or not.

1.9.5 Your Turn!

Component Overview: Students will work independently on practicing find the sum and/or difference of radical expressions. For this activity, release students to try these questions independently and bring students back together to review answers.



1.9.5 Your Turn!

1. Find the sum or difference. If you cannot add or subtract, circle the expression.

a. $\sqrt[3]{27000} - \sqrt[3]{512}$
22

b. $\sqrt{50} + \sqrt{18} + \sqrt{10}$
 $8\sqrt{2} + \sqrt{10}$

c. $\sqrt[3]{6} - \sqrt[3]{48}$
 $-\sqrt[3]{6}$

d. $32^{\frac{1}{2}} - 16^{\frac{1}{3}}$



Based on student performance in the prior activities, consider creating a small group for students who need differentiated support. With the small group, guide students through question 1a and have them do question 1c and 1d. If time permits, walk students through question part b. The goal of this group is to reteach the content for mastery, not necessarily to complete all problems.



Consider calling students to the board to show their work for each question. Allow students to provide feedback to their peers and agree or disagree with the final answers and process used to get there.



For a challenge, allow students to create an addition or subtraction problem and trade it with a partner, if time permits. Have the partner find the sum or difference and compare their answer to the author of the original problem and discuss the answer.

1.9.6 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door that assesses student understanding of finding the sum of a radical expression. This activity should be done independently to gather the most accurate data. Students should not use a calculator to complete the Wrap-Up.



1.9.6 Wrap-Up: Ticket Out the Door

- Find the sum.

$$17\sqrt{8} + 5\sqrt{8}$$

$$44\sqrt{2}$$



Student performance on this Wrap-Up can provide data that indicates additional support and differentiation that may be necessary.

Pay attention to what students put as their final answer. For example, if students put $22\sqrt{40}$, the student misunderstands the “like terms” component of this type of problem.